

Raina Wu

✉ rwu986@mit.edu

📍 Cambridge, MA

🌐 sleepypandu.github.io

🔗 [sleepypandu](#)

📄 [raina-wu-890758326](#)

Education

B.S. Massachusetts Institute of Technology, Computer Science (Course 6-4, Artificial Intelligence and Decision Making)

Cambridge, MA
Sept 2024 – May 2027

- GPA 5.0 / 5.0. Expected May 2027.
- **ML / AI:** 6.7960 Deep Learning, 6.7920 Reinforcement Learning: Foundations and Methods, 6.8300 Advances in Computer Vision, 6.7800 Inference and Information, 6.S184 Generative AI / Diffusion Lab (IAP); in progress: 6.8611 Natural Language Processing, 6.5940 TinyML and Efficient Deep Learning Computing, 6.4210 Robotic Manipulation
- **Theory and math:** 6.1220 Design and Analysis of Algorithms, 18.404 Theory of Computation, 18.701 Algebra I, 18.702 Algebra II, 18.510 Introduction to Mathematical Logic and Set Theory, 18.03 Differential Equations
- **Systems:** 6.1800 Computer Systems Engineering, 6.1910 Computation Structures, 6.1903 Introduction to Programming in C and Assembly, 6.1010 Fundamentals of Programming; in progress: 6.4400 Computer Graphics
- **Other:** 6.4590 Foundations of Information Policy, 24.09 Minds and Machines, 14.01 Principles of Microeconomics, 21H.152 Modern China, 24.02 Moral Problems and the Good Life

Experience

Hudson River Trading, Algo Dev Intern (incoming)

New York, NY
June 2027 – Aug 2027

Applied Compute, Research Intern (incoming)

Jan 2027 – May 2027

IBM Research, Research Intern — RL for LLM reasoning

June 2026 – Aug 2026

- Ran ablations and reproductions of existing self-distillation pipelines and uncovered failure modes of OPSD and SDPO.
- Ran a prefix-advantage study across Qwen3-1.7B/4B/8B characterizing where privileged-teacher signal localizes in a rollout. Submitted to the NeurIPS 2026 workshop on Transitioning from Pre-Training to Post-Training.
- Investigating other uses of distillation — whether privileged information can be used in a helpful, non-cheating way.

Hudson River Trading, WITI Intern — quantitative finance

New York, NY
Jan 2026 – Jan 2026

MIT-IBM Watson AI Lab, Researcher — RL for generative models

Cambridge, MA
May 2025 – May 2026

- Investigated RL fine-tuning of flow models for molecule and crystal generation.
- Co-author on *Tailored primitive initialization is the secret key to reinforcement learning* (ACL 2026); built the embedding-similarity pipeline used for the paper's diversity analysis, showing Tailor's warm-start data is more reasoning-diverse than the Standard-CoT and 4-STaR baselines.

Talkdesk, Software Engineering Intern — multi-agent systems

June 2025 – Aug 2025

- Prototyped agentic user memory for a multi-agent system and a multi-agent evaluation framework for a production RAG pipeline.

MIT DeCoDE Lab, Researcher — generative AI for design

Cambridge, MA
Jan 2025 – Jan 2025

- Built the sketch-to-3D retrieval pipeline (2D sketch → nearest 3D car mesh) and benchmarked retrieval models and similarity metrics; co-author on the resulting ASME IDETC-CIE 2025 paper.

MIT Media Lab, Researcher — AI security & agents

Cambridge, MA
Sept 2024 – May 2025

- Built attestation for AI agents negotiating inside GPU-backed trusted execution environments; wrote circom circuits producing zkSNARK proofs that the negotiation ran as specified.
- Built tool-using agents (Gmail, Google Calendar, Shopify APIs) and an evaluation harness for multi-agent negotiation.

- Proposed and evaluated an algorithm combining path-based link prediction with influence-maximization heuristics to identify future vital nodes in weighted collaboration networks.
- Introduced RA-2 and quasi-local RA-2, new link-prediction similarity metrics; built the simulation and evaluation pipeline (7 link-prediction metrics $\times$ 12 centrality measures $\times$ 8 selection algorithms, under simple and complex contagion); wrote the initial manuscript.
- Co-author (alphabetical order) on papers in *Physica A* (2026) and on arXiv (2024).

Publications

Tailored primitive initialization is the secret key to reinforcement learning

Aug 2026

Y. Yao, G. Zeng, R. Wu, Y. Zhang, D. Zhao, Z.-W. Hong, C. Gan

doi.org/10.18653/v1/2026.acl-long.1537 (ACL 2026 (Volume 1, Long Papers), pp. 33300–33318)

Influence prediction in collaboration networks: An empirical study on arXiv

2026

M. Lin, L. P. Schaposnik, R. Wu

doi.org/10.1016/j.physa.2026.131451 (Physica A: Statistical Mechanics and its Applications, 689, 131451)

AI agents in engineering design: A multi-agent framework for aesthetic and aerodynamic car design

Aug 2025

M. Elrefaie, J. Qian, R. Wu, Q. Chen, A. Dai, F. Ahmed

doi.org/10.1115/DETC2025-169682 (ASME 2025 IDETC-CIE, Volume 3B, 51st Design Automation Conference)

The social sphere model: Heuristic influence prediction in evolving networks

Feb 2024

M. Lin, L. P. Schaposnik, R. Wu

arxiv.org/abs/2402.03522 (arXiv preprint arXiv:2402.03522)

Honors

- **Putnam Mathematical Competition** — Top 220 (2025), Top 300 (2026)
- **High school:** Mathematical Olympiad Program (MOP) 2022, national top 60; USAJMO winner; USAMO qualifier; Math Prize for Girls national top 12 (2023)

Teaching

- **Teaching Assistant, 18.404 Theory of Computation** — MIT, Fall 2026 (current)
- **Teaching Assistant, Athemath** — 2022–2024
- **Curriculum Developer, AlphaStar Academy** — 2021–2022

Activities

- **MIT Syncopasian** — a cappella group (syncopasian.com)

Skills

Programming

Python, C++, Java, SQL

ML & Systems

RL for LLMs (PPO, GRPO, OPD, self-distillation), verl, vLLM, LoRA, FSDP, PyTorch, NumPy, multi-node GPU training (LSF), Docker, PostgreSQL, agentic systems, embedding calculations, circom / zkSNARKs

Languages

English (native), Mandarin Chinese (fluent)